

- 1 Solve the equation $4 \times 2^{x+2} - 5 \times 2^{2-x} = 3$. Give your answer correct to 3 significant figures. [4]

$$u = 2^x$$

$$(u > 0) \quad 4(2^2 \cdot u) - 5(2^2 \cdot \frac{1}{u}) = 3$$

$$16u - 20 \frac{1}{u} = 3$$

$$16u^2 - 3u - 20 = 0$$

$$u_{1,2} = \frac{3 \pm \sqrt{3^2 + 4(16)(20)}}{2(16)} \rightarrow \frac{3 + \sqrt{1289}}{32}$$

$$\rightarrow \frac{3 - \sqrt{1289}}{32} \text{ reject}$$

$$2^x = \frac{3 + \sqrt{1289}}{32}$$

$$\rightarrow x = 0.282 \text{ only}$$

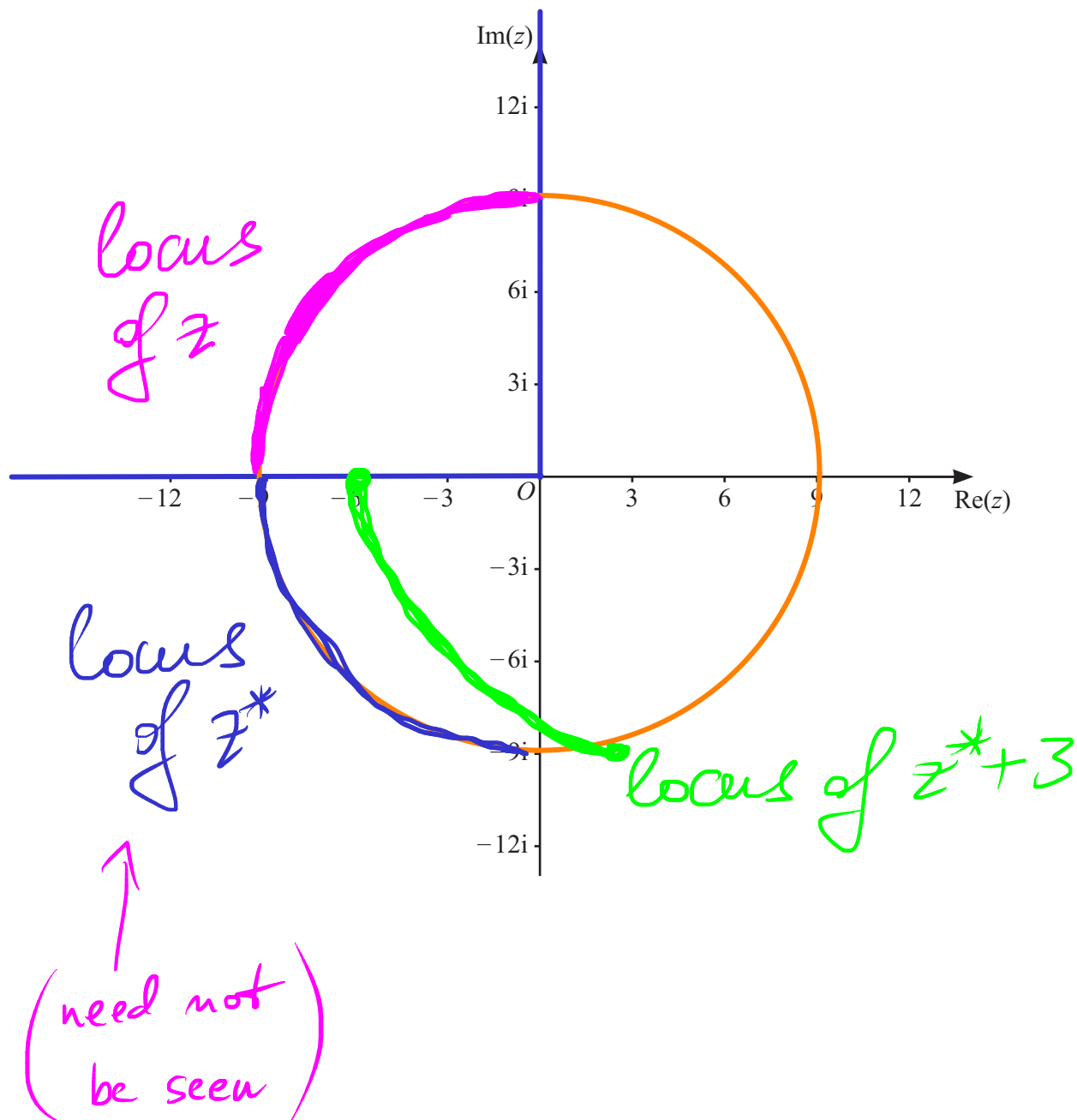
(Method of solving quadratic must be seen.)



- 2 The complex number z satisfies $|z| = 9$ and $\frac{1}{2}\pi \leq \arg z \leq \pi$.

(a) On the Argand diagram, sketch the locus of the points representing z . [2]

(b) On the **same diagram**, sketch the locus of the points representing $z^* + 3$. [2]



$$\frac{d}{dx} \sin^3 x = 3 \sin^2 x \cos x$$

- 3 Find the exact value of $\int_{\frac{1}{4}\pi}^{\frac{1}{3}\pi} 3 \sin x \sin 2x \, dx$.

Give your answer in the form $p\sqrt{3} + q\sqrt{2}$, where p and q are rational.

[4]

$$= \int_{\frac{1}{4}\pi}^{\frac{1}{3}\pi} 6 \sin^2 x \cos x \, dx = 2 \left[\sin^3 x \right]_{\frac{1}{4}\pi}^{\frac{1}{3}\pi}$$

$$= 2 \left(\left(\sin \left(\frac{\pi}{3} \right) \right)^3 - \left(\sin \left(\frac{\pi}{4} \right) \right)^3 \right)$$

$$= 2 \left(\left(\frac{\sqrt{3}}{2} \right)^3 - \left(\frac{\sqrt{2}}{2} \right)^3 \right) = 2 \left(\frac{3\sqrt{3}}{8} - \frac{\sqrt{2}}{4} \right)$$

$$= \frac{3}{4} \sqrt{3} - \frac{1}{2} \sqrt{2}$$

(Integrate by any
valid method)

Substitution
must be
seen



- 4 The coefficient of x^3 in the expansion of $(1-ax)^{\frac{2}{5}}$ is 1.

(a) Find the value of a .

[2]

$$\frac{\left(\frac{2}{5}\right)\left(\frac{-3}{5}\right)\left(\frac{-8}{5}\right)}{3 \cdot 2 \cdot 1} (-a)^3 = 1$$

$$a = -\frac{5}{2}$$

- (b) Hence, find the coefficient of x^4 in the expansion of $(2x+1)(1-ax)^{\frac{2}{5}}$.

[3]

$$\frac{\left(\frac{2}{5}\right)\left(\frac{-3}{5}\right)\left(\frac{-8}{5}\right)\left(\frac{-13}{5}\right)}{4 \cdot 3 \cdot 2 \cdot 1} (-a)^4 = -\frac{26}{625}a^4$$

$$= -\frac{13}{8}$$

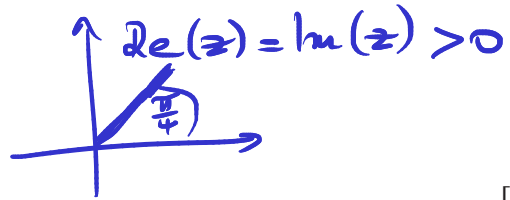
$$2(1) + 1\left(-\frac{13}{8}\right) = \frac{3}{8}$$

- (c) State the set of values of x for which the expansion in 4(b) is valid.

[1]

$$|-ax| < 1 \rightarrow \left|-\frac{5}{2}x\right| < 1 \rightarrow |x| < \frac{2}{5}$$

$$\rightarrow -\frac{2}{5} < x < \frac{2}{5}$$



- 5 It is given that $z = \frac{3+\lambda i}{\lambda+2i}$, where λ is a real constant.

(a) Find the value of λ for which $\arg z = \frac{1}{4}\pi$. [4]

$$\frac{3+\lambda i}{\lambda+2i} \cdot \frac{\lambda-2i}{\lambda-2i} = \frac{(3\lambda+2\lambda) + (\lambda^2-6)i}{\lambda^2+4}$$

$$5\lambda = \lambda^2 - 6$$

$$0 = \lambda^2 - 5\lambda - 6$$

$$\lambda_{1,2} = \frac{5 \pm \sqrt{5^2 + 4(1)(6)}}{2(1)} \rightarrow \lambda_1 = 6$$

$$\rightarrow \lambda_2 = -1 \text{ reject}$$

$$\rightarrow \lambda = 6 \text{ only}$$

(Method of solving the quadratic must be seen. Must reject the root -1 clearly.)

- (b) When λ has the value found in 5(a), find the exact value of $|z|$, making your method clear. [2]

$$z = \frac{30 + (36-6)i}{36+4}$$

$$= \frac{1}{40} \sqrt{30^2 + 30^2} = \frac{3}{4} \sqrt{2}$$



- 6 The polynomial $2x^4 + ax^3 + 4x^2 + bx - 3$ is denoted by $p(x)$.

It is given that $(x^2 + x + 1)$ is a factor of $p(x)$.

- (a) Find the values of a and b .

[4]

$$(x^2 + x + 1)(2x^2 + \alpha x - 3) = 2x^4 + ax^3 + 4x^2 + bx - 3$$

$$x^2: (1)(-3) + (1)(\alpha) + 2 = 4 \rightarrow \alpha = 5$$

$$x^3: (1)(\alpha) + (1)(2) = a = 7$$

$$x: (1)(-3) + (1)(\alpha) = b = 2$$

(Can also do long division)

- (b) Hence show that $(x + 3)$ is a factor of $p(x)$.

[2]

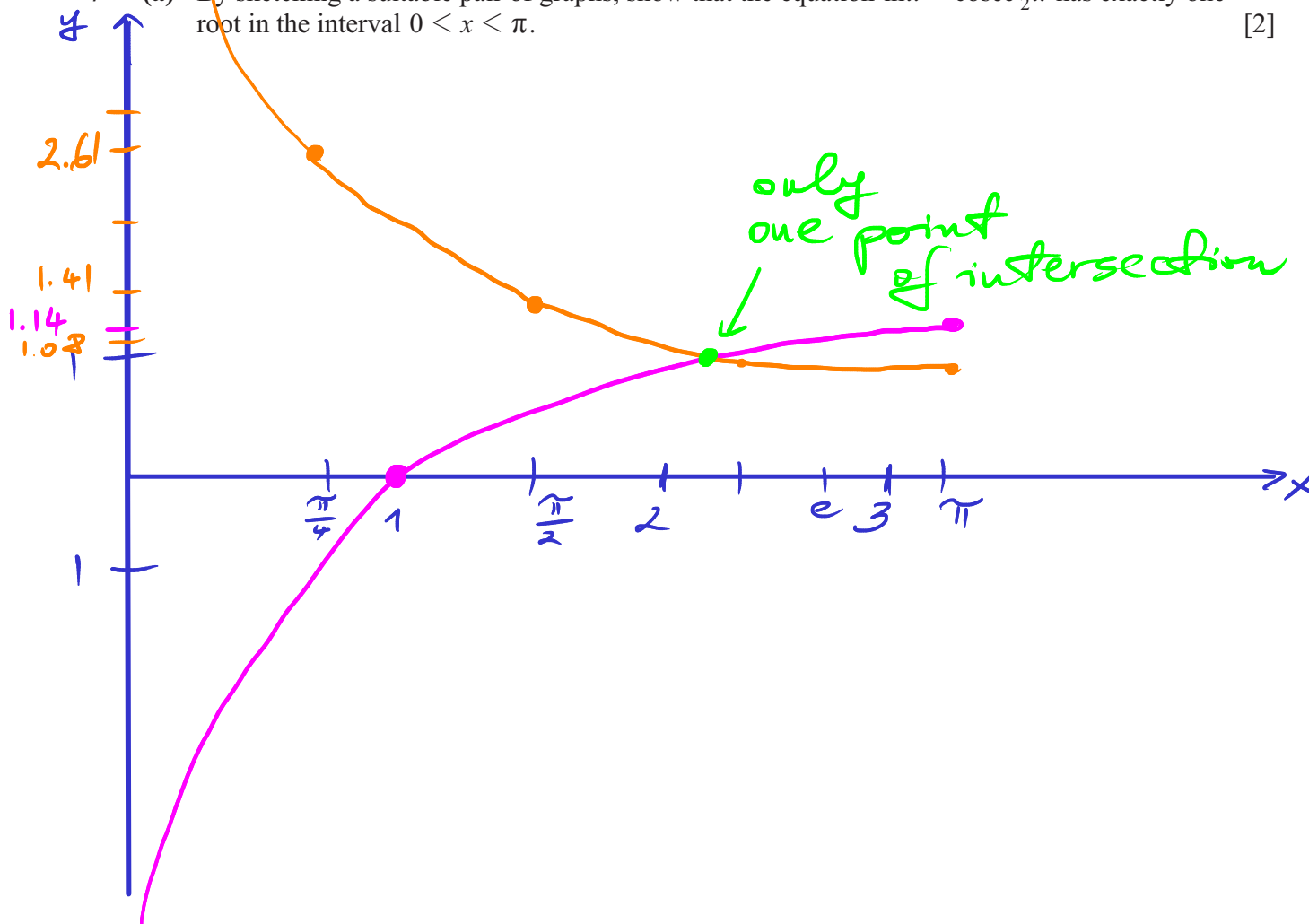
$$2x^2 + 5x - 3 \text{ at } x = -3 \text{ is}$$

$$2(-3)^2 + 5(-3) - 3 = 0$$

so $(x + 3)$ is a factor of $2x^2 + 5x - 3$
and thus of $p(x)$

(Can also substitute in $p(x)$)

- 7 (a) By sketching a suitable pair of graphs, show that the equation $\ln x = \operatorname{cosec} \frac{1}{2}x$ has exactly one root in the interval $0 < x < \pi$. [2]



- (b) Verify by calculation that this root lies between 2.6 and 2.9. [2]

$$\ln(2.6) - \operatorname{cosec}\left(\frac{1}{2}(2.6)\right) = -0.0823$$

$$\ln(2.9) - \operatorname{cosec}\left(\frac{1}{2}(2.9)\right) = 0.0574$$

there is a sign change in $f(x) = \ln(x) - \operatorname{cosec}\left(\frac{1}{2}x\right)$ between $x = 2.6$ and $x = 2.9$, and on $2.6 < x < 2.9$ $f(x)$ is continuous so there must be a root in $2.6 < x < 2.9$

(Must answer with a full sentence)



- (c) Use the iterative formula $x_{n+1} = \exp\left(\operatorname{cosec} \frac{1}{2} x_n\right)$ to determine the root correct to 3 decimal places. Give the result of each iteration to 5 decimal places. [3]

[$\exp(x)$ is an alternative notation for e^x .]

$$x_1 = 2.6$$

$$x_2 = \exp\left(\operatorname{cosec} \left(\frac{1}{2}(2.6)\right)\right) = 2.82306$$

$$x_3 = \exp\left(\operatorname{cosec} \left(\frac{1}{2}(2.82306)\right)\right) = 2.75335$$

$$x_4 = 2.77081$$

$$x_5 = 2.76609$$

$$x_6 = 2.76734$$

$$x_7 = 2.76700$$

$$x_8 = 2.76710$$

$$x_9 = 2.76707$$

$$x_{10} = 2.76708$$

(Must show at least TWO iterations with substitution)

$$\text{so } x = 2.767$$

(Must do enough iterations for x_n to repeat to 4dp)

- 8 The variables x and y satisfy the differential equation

$$ye^{3x} \frac{dy}{dx} = x(y+5).$$

It is given that $y = 0$ when $x = 0$.

Solve the differential equation to obtain an equation in x and y .

[8]

$$\int \frac{y}{y+5} dy = \int x e^{-3x} dx$$

$$\int \frac{y+5-5}{y+5} dy = \int \left(1 - \frac{5}{y+5}\right) dy =$$

$$= y - 5 \ln|y+5| + C = \int x e^{-3x} dx$$

<table border="0" style="margin-left: auto; margin-right: auto;"> <tr> <td style="text-align: center;">D</td> <td style="text-align: center;">I</td> </tr> <tr> <td style="text-align: center;">+ x</td> <td style="text-align: center;">e^{-3x}</td> </tr> <tr> <td style="text-align: center;">- 1</td> <td style="text-align: center;">$-\frac{1}{3}e^{-3x}$</td> </tr> </table>	D	I	+ x	e^{-3x}	- 1	$-\frac{1}{3}e^{-3x}$	$= -\frac{1}{3}x e^{-3x} + \frac{1}{3} \int e^{-3x} dx$ $= -\frac{1}{3}x e^{-3x} - \frac{1}{9} e^{-3x}$
D	I						
+ x	e^{-3x}						
- 1	$-\frac{1}{3}e^{-3x}$						

at $x=0, y:0$: $0 - 5 \ln|5| + C = 0 - \frac{1}{9} e^0$

$$C = 5 \ln 5 - \frac{1}{9}$$

$$y - 5 \ln|y+5| + 5 \ln 5 - \frac{1}{9} = -\frac{1}{3}x e^{-3x} - \frac{1}{9} e^{-3x}$$

(Can use other methods)

9 Let $I = \int_1^3 \frac{x^3}{3+x^2} dx$.

(a) Using the substitution $x = \sqrt{3} \tan u$, show that $I = \int_{\frac{1}{6}\pi}^{\frac{1}{3}\pi} 3 \tan^3 u \, du$. [4]

$$\frac{dx}{du} = \sqrt{3} \sec^2 u \rightarrow dx = \sqrt{3} \sec^2 u \cdot du$$

$$u = \tan^{-1}\left(\frac{x}{\sqrt{3}}\right)$$

$$x=1 \rightarrow u = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$$

$$x=3 \rightarrow u = \tan^{-1}\left(\frac{3}{\sqrt{3}}\right) = \frac{\pi}{3}$$

$$I = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{(\sqrt{3} \tan u)^3}{3 + (\sqrt{3} \tan u)^2} \cdot \sqrt{3} \sec^2 u \, du$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{3\sqrt{3} \tan^3 u}{3(1 + \tan^2 u)} \cdot \sqrt{3} \sec^2 u \, du$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{3}{3} (\sqrt{3})(\sqrt{3}) \tan^3 u \frac{\sec^2 u}{\sec^2 u} \, du$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} 3 \tan^3 u \, du$$

(Any fully complete method)



- (b) Hence, or otherwise, find the exact value of I . Give your answer in the form $p + q \ln r$, where p, q and r are rational. [5]

$$\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} 3 \tan^3 u \, du = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} 3 \tan u (\sec^2 u - 1) \, du = (*)$$

$$\frac{d}{du} (\tan^2 u) = 2 \tan u (\sec^2 u) \quad \text{(Use any method to integrate)}$$

$$\frac{d}{du} \ln |\cos u| = \frac{-\sin u}{\cos u} = -\tan u$$

$$(*) = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} 3 \tan u \sec^2 u \, du - \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} 3 \tan u \, du$$

$$= \left[\frac{3}{2} \tan^2 u + 3 \ln |\cos u| \right]_{\frac{\pi}{6}}^{\frac{\pi}{3}}$$

$$= \frac{3}{2} \tan^2 \left(\frac{\pi}{3} \right) + 3 \ln \left| \cos \left(\frac{\pi}{3} \right) \right|$$

$$- \frac{3}{2} \tan^2 \left(\frac{\pi}{6} \right) - 3 \ln \left| \cos \left(\frac{\pi}{6} \right) \right|$$

$$= \frac{3}{2} (3) + 3 \ln \left(\frac{1}{2} \right) - \frac{3}{2} \left(\frac{1}{3} \right) - 3 \ln \left(\frac{\sqrt{3}}{2} \right)$$

$$= 4 - 3 \ln 2 - 3 \cdot \frac{1}{2} \ln 3 + 3 \ln 2$$

$$= 4 - \frac{3}{2} \ln 3$$

or equivalent
simplified
in given
form

- 10 The variables x and y satisfy the equation $y^2 = k \frac{x-2}{x+2}$, where k is a constant.

(a) Show that $\frac{dy}{dx} = \frac{2y}{x^2-4}$.

[5]

$$\begin{aligned} \frac{d}{dx} y^2 &= 2y \cdot \frac{dy}{dx} \\ &= \frac{d}{dx} k \left(\frac{x-2}{x+2} \right) = k \frac{(1)(x+2) - (x-2)(1)}{(x+2)^2} \end{aligned}$$

$$\frac{dy}{dx} = \frac{k \frac{x+2-x+2}{(x+2)^2}}{2y} = \frac{4k}{2y(x+2)^2}$$

$$= \frac{(4k)(2y)}{4y^2(x+2)^2} = \frac{(4k)(2y)}{4k \left(\frac{x-2}{x+2} \right) (x+2)^2}$$

$$= \frac{2y}{(x-2)(x+2)} = \frac{2y}{x^2-4}$$

(Can also use partial fractions

$$\frac{x-2}{x+2} = 1 - \frac{4}{x+2} \text{ before}$$

taking derivative instead
of the quotient rule



(b) Given that $k = 5$, find the angle between the tangents to the curve when $x = 3$.

Give your answer in the form $a \tan^{-1}\left(\frac{b}{c}\right)$, where a , b and c are integers.

[4]

$$y^2 = 5 \frac{(3-2)}{(3+2)} = 1 \rightarrow y_1 = 1 \quad y_2 = -1$$

$$\text{at } y_1 = 1: \frac{dy}{dx} = \frac{2(1)}{(3)^2 - 4} = \frac{2}{5}$$

$$\text{at } y_2 = -1: \frac{dy}{dx} = \frac{2(-1)}{(3)^2 - 4} = -\frac{2}{5}$$

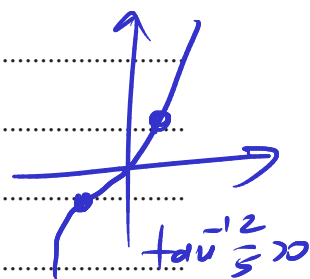
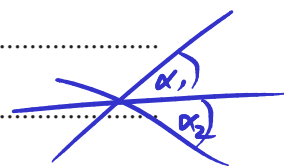
angle between =

$$\left| \tan^{-1}\left(\frac{2}{5}\right) - \tan^{-1}\left(-\frac{2}{5}\right) \right|$$

$$= \left| \tan^{-1}\left(\frac{2}{5}\right) + \tan^{-1}\left(\frac{2}{5}\right) \right|$$

$$= 2 \tan^{-1}\left(\frac{2}{5}\right)$$

(Final answer must not have absolute value in it)





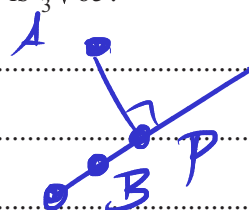
$$16 \quad A = \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix} \quad B = \begin{pmatrix} 4 \\ 2 \\ 4 \end{pmatrix}$$

- 11 The points A and B have position vectors $\vec{OA} = 2\mathbf{i} + 2\mathbf{j} - \mathbf{k}$ and $\vec{OB} = 4\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$ relative to the origin, O .

- (a) Show that the perpendicular distance from A to the line through O and B is $\frac{1}{3}\sqrt{65}$. [5]

$$P = \mu B = \begin{pmatrix} 4\mu \\ 2\mu \\ 4\mu \end{pmatrix}$$

$$\vec{OB} \perp \vec{AP} = \begin{pmatrix} 4\mu - 2 \\ 2\mu - 2 \\ 4\mu + 1 \end{pmatrix}$$



$$(4)(4\mu - 2) + (2)(2\mu - 2) + (4)(4\mu + 1) = 0$$

$$\mu(16 + 4 + 16) = 8 + 4 - 4$$

$$\mu = \frac{8 + 4 - 4}{16 + 4 + 16} = \frac{2}{9}$$

$$|\vec{AP}| = \sqrt{\left(4\left(\frac{2}{9}\right) - 2\right)^2 + \left(2\left(\frac{2}{9}\right) - 2\right)^2 + \left(4\left(\frac{2}{9}\right) + 1\right)^2}$$

$$= \sqrt{\frac{65}{9}} = \frac{1}{3}\sqrt{65}$$

(Can also do $\vec{OP} \perp \vec{AP}$ then reject $\mu = 0$)



$$A = \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix} \quad B = \begin{pmatrix} 4 \\ 2 \\ 4 \end{pmatrix}$$

- (b) The point C has position vector $\vec{OC} = 3\mathbf{i} + p\mathbf{j} + q\mathbf{k}$, where p and q are constants.

Given that

- $\vec{OC}$ is perpendicular to $\vec{AB}$ and
- angle $AOC = \text{angle } COB$,

find the values of p and q .

[7]

$$\vec{AB} = \begin{pmatrix} 4-2 \\ 2-2 \\ 4-(-1) \end{pmatrix} = \begin{pmatrix} 2 \\ 0 \\ 5 \end{pmatrix}$$

$$0 = \vec{OC} \cdot \vec{AB} = 3(2) + 0 + 5q$$

$$q = \frac{-3(2)}{5} = -\frac{6}{5}$$

$$\vec{OA} \cdot \vec{OC} = \vec{OC} \cdot \vec{OB}$$

$$|\vec{OA}| |\vec{OC}| = |\vec{OC}| |\vec{OB}|$$

$$\frac{(2)(3) + 2p - q}{\sqrt{2^2 + 2^2 + 1^2}} = \frac{4(3) + 2p + 4q}{\sqrt{4^2 + 2^2 + 4^2}}$$

$$\frac{2p + \frac{36}{5}}{3} = \frac{2p + \frac{36}{5}}{6}$$

$$2p + \frac{36}{5} = 0$$

$$p = \frac{-\frac{36}{5}}{2} = -\frac{18}{5}$$